Fire-aware robot dogs: machine-learning wildfire forecasts for safe navigation and scouting

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Legged robots could scout a wildfire edge in place of firefighters, but only if they know where the fire will be. We combine a machine-learning spread forecast with a terrain-aware route planner that keeps a robot dog ahead of the predicted fire and holds an escape path home. In a simulated mission on real Big Sur terrain, an unforecast wind shift cut forecast agreement from 82% to 61%; the dog's field readings restored it to 80%, and the dog returned safely.

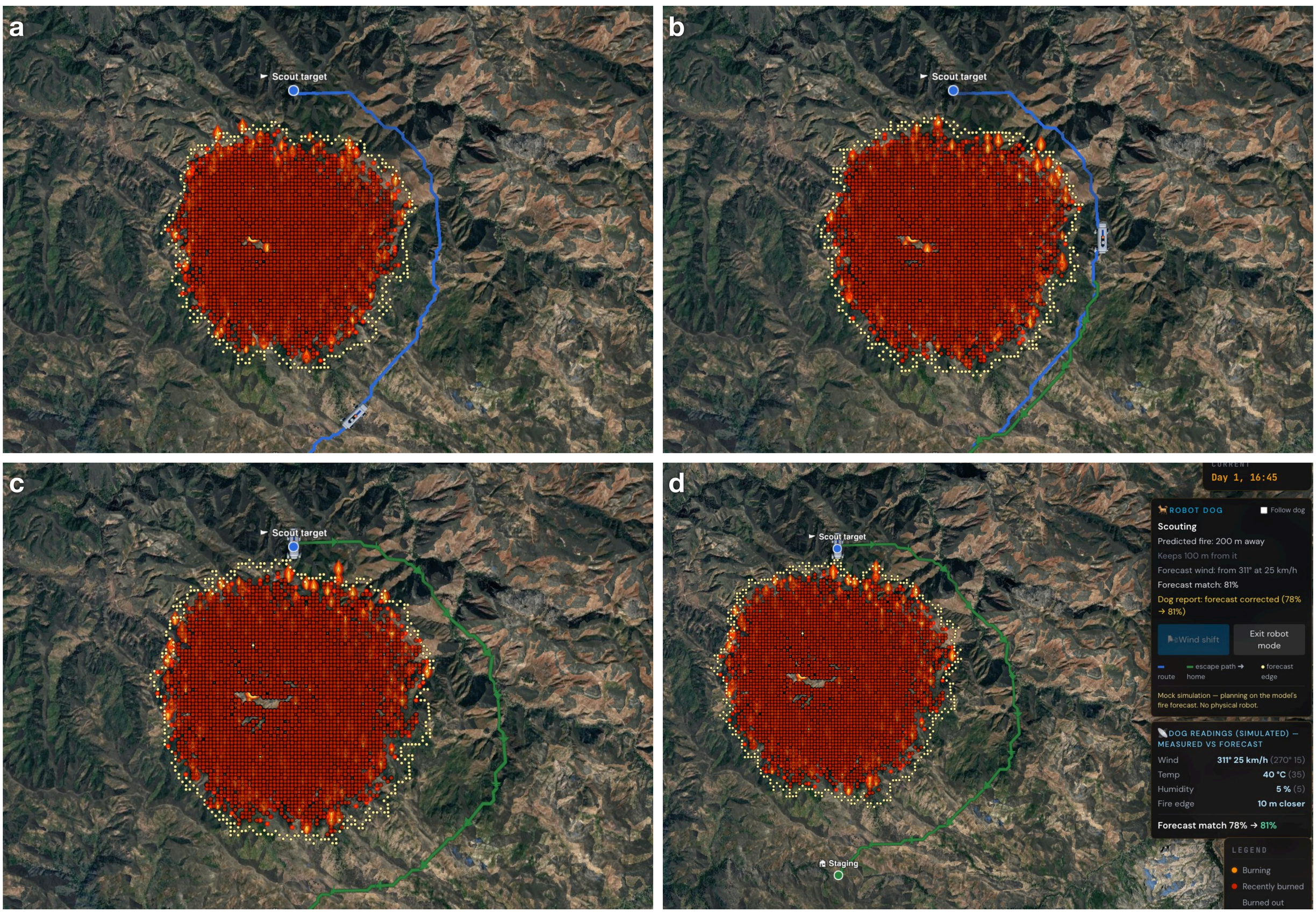


Fig. 3 | A simulated scouting mission on real terrain (Big Sur, California). **a**, At 15:00 the dog plans a route (blue) that reaches every cell at least 30 min before its forecast ignition time. **b**, At 15:30 the wind shifts; the hidden true fire (red) departs from the forecast edge (yellow dots) while the escape path (green) stays clear. **c**, The dog re-plans on the corrected forecast and scouts the fire edge. **d**, Readings sent at 15:45 compared with the forecast's assumptions. Simulator screenshots; the robot and its sensors are simulated.

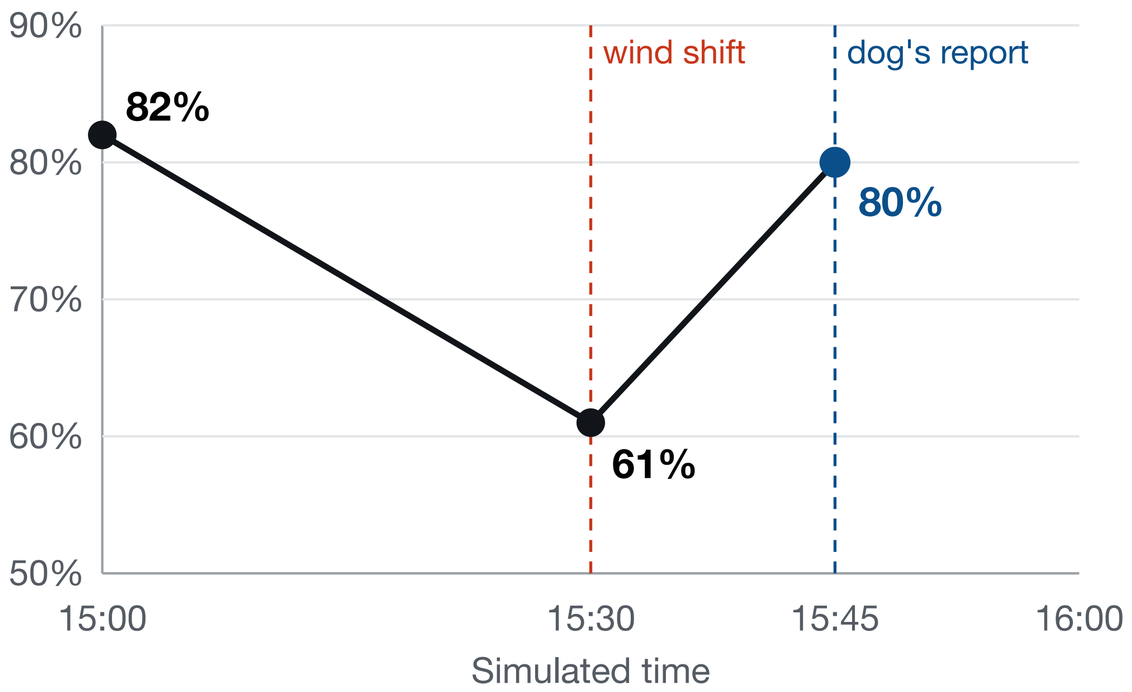


Fig. 4 | Forecast agreement with the true fire. Agreement between the forecast and the hidden true fire over the next 4 h of spread, at launch, after the wind shift, and after the dog's report (one mission).

Table 2 | Field readings versus forecast assumptions at the dog's report (15:45).

Quantity	Measured	Forecast
Wind direction	45°	270°
Wind speed	25 km/h	15 km/h
Temperature	40 °C	35 °C
Relative humidity	5%	5%
Fire-edge position	20 m closer than forecast	

The correction raised agreement from 61% to 80% (Fig. 4). The dog then re-planned, scouted the edge and returned to staging at about 20:00, before the fire arrived.